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Review Article

Phantom auditory perception (tinnitus): mechanisms of generation and perception

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SUMMARY

Phantom auditory perception – tinnitus – is a symptom of many pathologies. Although there are a number of theories postulating certain mechanisms of its generation, none have been proven yet. This paper analyses the phenomenon of tinnitus from the point of view of general neurophysiology. Existing theories and their extrapolation are presented, together with some new potential mechanisms of tinnitus generation, encompassing the involvement of calcium and calcium channels in cochlear function, with implications for malfunction and aging of the auditory and vestibular systems.

It is hypothesized that most tinnitus results from the perception of abnormal activity, defined as activity which cannot be induced by any combination of external sounds. Moreover, it is hypothesized that signal recognition and classification circuits, working on holographic or neuronal network-like representation, are involved in the perception of tinnitus and are subject to plastic modification. Furthermore, it is proposed that all levels of the nervous system, to varying degrees, are involved in tinnitus manifestation. These concepts are used to unravel the inexplicable, unique features of tinnitus and its masking. Some clinical implications of these theories are suggested.

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1. INTRODUCTION

1.1 *Epidemiology of phantom auditory perception*

Phantom auditory perception, described as tinnitus in the clinical literature, occurs with surprisingly high frequency and affects about 35% of the general population^{27,169,272}, with about 15% reporting its presence as frequent or continuous^{7,169}. Tinnitus has been described in association with nearly every form of ear pathology. It occurs in 85% of patients with ear problems^{26,169}, and afflicts nearly 36 million Americans, 7–9 million severely^{26,27,169,272}, with high levels of annoyance precluding normal life in about 1 million^{27,169,272}. Nevertheless, the pathophysiology of tinnitus has remained obscure and its treatment therefore elusive. With respect to a cure for tinnitus, the most recent clinical information^{27,56,89,248} is rather pessimistic. Although some relief has been observed in patients treated with auditory masking procedures^{93,94,169,272}, electrical stimulation^{8,92,141}, pharmacological and other methods^{27,38,40,166,169,243,271}, most approaches seem to act on the patient's attitude toward their tinnitus rather than on the underlying mechanisms^{94,199,274}. Additionally, a strong non-specific placebo effect has been observed, making simple evaluation of treatment effectiveness open for discussion^{27,43,48,51,165,254}. For most sufferers, particularly the most disturbed and distressed, help remains inaccessible. Significant progress in this field can only be achieved by determining the mechanisms of different types of tinnitus generation. A better understanding of how neuronal activity related to tinnitus is processed within the nervous system is needed to comprehend possible mechanisms of tinnitus attenuation. Such information will eventually allow for mechanism-specific treatment of tinnitus patients.

1.2 *An animal model of tinnitus*

Until now, essentially all work on tinnitus was done on humans, with the consequential implications and restrictions thereby resulting, owing to the lack of a valid animal model of tinnitus. Recently, such a model was created in this laboratory^{111–116,118,121} and is presently being used to evaluate different possible mechanisms of tinnitus and methods of

its alleviation, as well as to address the general issues of perception and hallucinations¹¹⁷. This model implemented two independent strategies: (i) a behavioral paradigm, based on Pavlovian suppression technique; and (ii) an electrophysiological evaluation of the spontaneous activity of single cells recorded from the inferior colliculus.

The behavioral approach is based on the well-established observation that salicylates cause auditory phantom sensations systematically in humans^{61,169,182} and the following reasoning. It is impossible to induce tinnitus in a systematic manner for short, precisely controlled periods of time. This precludes utilization of typical behavioral techniques aimed at emphasis of the relation between short, well-defined sensory stimuli, which are stronger than background, and a reinforcement. During these procedures, animals are trained to habituate to the background. To enable tinnitus detection, it is necessary to reverse this situation and to utilize training which would enhance the importance of a continuous signal (tinnitus), constituting a portion of the background.

To achieve this goal, the following procedure was developed as a modification of the approach created by Estes and Skinner⁵². This paradigm is based on the creation of antagonism between ongoing, appetitively-based behavior and aversively-based response elicited by Pavlovian training, confronting two opposite motivations: thirst and fear. The animals' continuous reactions (licking reinforced by water) are partially attenuated during the conditioned stimulus by fear, which has been previously associated with the conditioned stimulus by proper training. Since the animal licks numerous times throughout the duration of the conditioned stimulus (30 s), the thirst level is kept constant, and the balance between thirst and fear results in a different extent of reaction suppression, it is then possible to evaluate gradual changes in the animals' fear.

The main idea of the tinnitus detection paradigm was based on creating a situation in which the presence of a continuous auditory signal would be associated by the animal with safety and the absence of it (silence) with danger. To achieve this, pigmented rats were exposed 24 h/day to mild (62 dB SPL) noise and were trained to associate silence with danger by terminating 4 presentations of 30 s of noise offset with a single mild electric shock. Next, the animals went through a passive extinction period, when offset of noise was presented but without the shock. The expectation was that if during extinction the animals experienced salicylate-induced tinnitus (evoked by daily injection of sodium salicylate) while external noise was off, it would be interpreted as a substitute for background noise, hence as a signal of safety. Therefore offset of noise should suppress licking to a smaller extent, and the process of extinction should occur more rapidly than in the control group injected with saline. Conversely, if the animals were trained in a Pavlovian suppression situation when salicylate-induced tinnitus was present, the rats would be conditioned to the sound of tinnitus as the conditioned stimulus, since during offset of external noise, tinnitus would become the dominant signal. This group of animals would be expected to exhibit stronger and longer lasting suppression than the previously described groups. Thus, daily salicylate injection should have an opposite effect on the animals' behavior, depending on whether the drug was introduced before or after the day of suppression training.

The results obtained, presented, and discussed in detail elsewhere¹¹⁵ fully confirmed these expectations. Relative to control subjects injected with saline, subjects starting injections prior to training exhibited more severe suppression and persistence of suppression. Conversely, subjects beginning injections after training showed less suppression and rapid recovery of drinking.

The important question to be answered was if the observed effects were due to the action of salicylate on the auditory system or perhaps due to non-specific action of the

drug. To test this, another experiment was performed paralleling the procedure outlined above but switching to a visual modality, e.g. using continuous light as background, with offset of this light serving as the conditioned stimulus. In this situation, the effects of salicylate observed previously were eliminated, indicating that the salicylate-induced effects were specifically auditory. A number of other control experiments were performed, including mimicking tinnitus with continuous external tone, which showed the same effect as salicylate¹¹⁵; decreasing the level of background noise at the same stages as salicylate introduction, to evaluate the effect of salicylate-induced increase of hearing threshold, had no effect¹¹⁰.

Furthermore, we obtained analogous results using another drug, quinine¹¹⁶. Quinine presumably has a different mechanism of action on the auditory system than does salicylate^{215,249}, but shares with salicylate tinnitus-inducing properties in humans^{61,169}. These combined results support the postulate that this paradigm detects salicylate-induced phantom auditory perception in rats.

In an electrophysiological study of tinnitus, the spontaneous activity of inferior colliculus cells was analyzed before and after salicylate administration¹²¹. This approach was based on Evans' report^{53,54} which shows increased spontaneous activity of auditory nerve fibers and the appearance of abnormal patterns of activity after salicylate. Our results showed that the spontaneous activity of inferior colliculus cells increased after salicylate, while the spontaneous activity in the non-auditory part of the cerebellar vermis remained stable¹²¹. Additionally, the presence of abnormal activity was indicated. These findings support the hypothesis that tinnitus results from aberrant neural activity within the auditory pathways, which is interpreted as sound at higher auditory centers.

It was necessary to assure that the salicylate doses being used were inducing sufficient levels of the drug in perilymph to interfere with cochlear functions, and that the behavioral differences observed between groups of animals were not due to varying salicylate levels. Sampling of serum, cerebrospinal fluid, and perilymph at various time intervals after salicylate showed levels of the drug^{114,119} comparable with levels reported to alter cochlear transduction functions²¹⁵, to decrease the turgor of outer hair cells (OHC)²⁰, and to alter their membrane properties^{6,244}. Finally, different experimental treatments and paradigms were without effect on salicylate uptake, eliminating the possibility that the behavioral differences observed among groups were due to variations in salicylate levels.

Work to date has focused on developing and establishing an animal model of tinnitus using both behavioral and electrophysiological approaches. The behavioral part shows that animals are perceiving tinnitus in a given experimental situation, while the electrophysiology elucidates possible cochlear and neural mechanisms of tinnitus generation. We are now evaluating the hypothesis, presented in more detail further in this text, regarding the involvement of calcium and calcium channels in normal functions and dysfunctions of the cochlea^{111,112}.

1.3 Basic science connotation

Although virtually all data available to date were gathered clinically, the phenomenon of tinnitus is very interesting for basic science. It offers an interesting tool to study mechanisms of phantom perception in general and can be used as a starting point to study hallucinations. Analysis of the tinnitus phenomenon should provide better insight into the mechanisms of perception and the representation of external stimuli in the nervous system. The relation between a perception and the stimulus evoking it is one of the basic issues of psychological theory. Plasticity of the brain supports the possibility of phantom perception in humans, originating from the ideas of gestalt psychologists⁹⁷ and

continuing in modern theories of perception^{41,77}. With respect to auditory neuroscience, tinnitus offers the challenging question of how relatively weak, continuous signals are being discriminated from the background of spontaneous activity without undergoing habituation. Tinnitus resembles phantom somatosensory and phantom pain perceptions²⁶². Information obtained from tinnitus research might help to better understand these phenomena.

In the literature, the terms 'objective' and 'subjective' tinnitus are sometimes used with the meaning that the sound of 'objective' tinnitus can be detected by an observer. As tinnitus is by definition a subjective phantom sensation, use of the terms 'subjective' and 'objective' is not advisable. Some tinnitus patients may be experiencing somatosounds^{27,169}, the result of abnormal blood flow or myogenic activity, and these phenomena should be excluded and managed separately.

Furthermore, it is important to stress that tinnitus is a physiological disorder of the auditory system, not a psychological or psychiatric disorder^{65,169}. Unfortunately in the past, some tinnitus patients were directed to a psychiatrist for treatment of schizophrenia, for which auditory hallucinations are landmark^{24,95,292}. Analysis of the psychological profile of tinnitus patients reveals the incidence of depression, which might be due to tinnitus, but the general psychosomatic profile remains normal⁶⁵.

1.4 General aims

Basic, well-established concepts of neurophysiology are being used in this paper to analyze the possible mechanisms of tinnitus generation and perception, indicating implications and predictions arising from this approach. These concepts are used to explain the puzzles of tinnitus, and to propose certain clinical implications. I would like to stress that since the current information available on tinnitus was collected under clinical or psychoacoustical pretenses, the type of information that was gathered is not appropriate for evaluating the correctness of my theories. Therefore, some of my hypotheses regarding tinnitus perception arise from theoretical predictions. Although the available data are in agreement with my proposals, more information is needed for their proof. Nevertheless, the hypotheses presented here are testable and offer a theoretical, experimental and clinical framework for work on tinnitus.

2. TINNITUS GENERATION

Several general hypotheses describing the origin of sensorineural tinnitus have been proposed^{157,177,230,260,261}; however, none has been proven yet. There is consensus that tinnitus is the result of aberrant neural activity within the auditory pathways and that such activity is erroneously interpreted as sound by auditory centers, but the agreement ends here. Nevertheless, there is clearly a variety of different conceivable mechanisms of tinnitus generation.

The majority of tinnitus cases are being related to cochlear dysfunction¹⁶⁹. The organ of Corti represents a very complex and delicately balanced micromechanical system. The discovery by Russell and Sellick^{220,221} that hair cells have tuning curves as sharp as single auditory nerve fibers showed that there is no need to postulate neuronal sharpening of frequency response beyond the basilar membrane mechanics. This caused reconsideration of previous concepts concerning the micromechanical properties of the organ of Corti^{136,159,241,290,304}. The OHC play an essential role in the frequency selectivity of the basilar membrane^{37,219}. The rigid coupling of cilia to hair cells is crucial for their normal function^{85,260,261,301-304}. The cilia of hair cells are stiff and brittle^{60,256,257}, the rigidity of

their coupling with the hair cell membrane being dependent upon the concentration of intracochlear calcium¹⁹¹. Measurement of the kinetics and transfer functions of ciliary movements has shown a strong dependence of these functions on calcium concentration^{34,60,98,191}. All these findings can be used to propose mechanisms of tinnitus generation based on cochlear dysfunction.

2.1 *Spontaneous otoacoustic emission*

The involvement of active mechanical processes in tinnitus generation has been suggested^{88,132,286–288}. These active processes, even when responsible for generating acoustic energy within the cochlea, do not seem to be simplistically related to tinnitus as initially hoped^{54,132,286,287}. An analysis of the population of tinnitus sufferers revealed that there is little or no correlation between the subjective characteristics of tinnitus and spontaneous otoacoustic emission^{16,87,130,132,170,205,217,263,298,300}. Nevertheless, a correlation occurs in a small percentage, estimated at 4%, of all tinnitus cases²⁷. Recent results by Penner indicate that spontaneous otoacoustic emission might be responsible for tinnitus^{200–202}. Then salicylate, which is well known to induce tinnitus in humans^{61,169,182} and was reported to abolish spontaneous otoacoustic emission^{162,167,170} and to suppress active processes in the cochlea^{6,20,212,213,244}, should relieve tinnitus. This has been described only in one case when, in parallel with the disappearance of initial tinnitus, salicylate-induced tinnitus appeared²⁰². More data are needed to evaluate the significance of spontaneous otoacoustic emission related tinnitus. Nevertheless, active processes might be involved in tinnitus generation but in a more complex manner as described in the sections to follow.

2.2 *Stereocilia decoupling*

In 1980, Tonndorf suggested an alteration in the mechanical coupling between hair cells and the tectorial membrane may be the basis for the origin of tinnitus^{260,261}. His hypothesis was derived from work by Harris⁸⁵, who calculated the level of thermal noise at the input of hair cells. Harris concluded that loose coupling of hair cell cilia to the tectorial membrane should result in increased thermal noise of the system, up to 30 dB above the threshold of sound perception. Tonndorf deduced that such an increase of thermal noise will be perceived as tinnitus. Due to the frequency-specific properties of segments of the basilar membrane, depending on the extent of basilar membrane involved, perception of the resulting noise varies from broad-band to very-narrow-band, the latter sounding essentially like pure tone. This theory predicts the appearance of center clipping of an auditory signal, resulting in poor speech perception and recruitment, phenomena observed in tinnitus patients^{27,169,260,261}. To my knowledge, this hypothesis has not been confirmed experimentally.

It is possible to extrapolate this theory further and to propose other mechanisms of decoupling the tectorial membrane from the basilar membrane. Mechanical decoupling might occur not only at the tectorial membrane but also at the attachment of the cilia to the OHC. Decreased strength of rootlet attachments in the apical plate of OHC would introduce 'free play' of the cilia and should have the same decoupling effect as in Tonndorf's model, resulting in increased thermal noise. Interestingly, exposure to intense sound, which is known as a method to produce temporary tinnitus in humans, causes bending and disorganization of the cilia, followed by damage to the rootlets as the first morphological findings observed after noise exposure^{150,156,232–235}. Such damage will result in mechanical decoupling of the tectorial membrane from the OHC system.

2.3 *Discordant damage of inner and outer hair cells*

Accumulated data show that a traumatic agent, such as noise or ototoxic drugs, causes cochlear damage starting its action on the basal high-frequency part of the basilar membrane, with OHC affected first, followed by damage to inner hair cells (IHC)^{29,103,154–156,234,253}. Therefore on a partially damaged basilar membrane, there will be some areas with totally damaged OHC and IHC, and other regions with damaged OHC while IHC remain reasonably intact. This last area is of particular interest. The mechanical properties of the organ of Corti will be seriously affected, due to disintegration of the coupling between the basilar and tectorial membranes, while IHC will be sufficiently functional. It is possible to expect abnormal movements of the basilar membrane and perhaps local collapse of the tectorial membrane, decreasing the distance between IHC cilia and the tectorial membrane to the extent of physical contact and bending of the cilia. This might cause tonic depolarization of IHC, resulting in the appearance of abnormal activity in afferent fibers. The hypothesis outlined here was proposed by Stypulkowski²⁵³ and might be further extended as follows to include possible loops involving efferent innervation of OHC and IHC systems.

It has been proposed that the position of the basilar membrane is adjusted to provide an optimal working point for transduction characteristics of the organ of Corti by changing the length of OHC under the influence of the efferent system innervating OHC^{147,148,296}. Afferents coming from OHC should provide information describing their working point which subsequently, after processing within the brainstem, will be fed back to efferent fibers to adjust the length of OHC. Decrease or lack of input from OHC belonging to a portion of the basilar membrane might result in decreased activity within efferent fibers¹⁸, aimed at increased negative dampening (through OHC) and decreased inhibition on afferents coming from IHC. This will yield enhanced activation of normal IHC, resulting in abnormal activity perceived as tinnitus. Since an efferent fiber branches profoundly in the cochlea, reaching as many as 100 OHC over a distance of up to 3 mm of basilar membrane¹⁵³, this will cause a release from inhibition of OHC and IHC from the neighboring normal portion of the basilar membrane. The activity due to this phenomenon can be further enhanced by edge effect, as described subsequently.

The above hypothesis can be used to explain several mysteries of tinnitus, such as: (i) why the pitch of tinnitus usually correlates with the slope of the audiogram, a common clinical observation⁸⁸; (ii) why patients with the same audiograms might or might not have tinnitus; (iii) the mechanisms of noise-induced temporary tinnitus. Moreover, it points out a new approach for evaluation of the tinnitus patient.

The slope of the audiogram corresponds to the area on the basilar membrane where partial damage of IHC and OHC occurs. Hence this is exactly the area where one can expect to find a portion of the basilar membrane with functional IHC and damaged OHC. The difference among patients with seemingly identical hearing loss represents a different extent of OHC and IHC loss⁸². It has been shown that exposure to the same noise might result in dissimilar patterns of cochlear damage in different subjects^{39,103,158}, and that the diffuse loss of OHC up to 30% might not have a significant impact on hearing threshold^{15,82,225}. This observation further supports the postulate that hearing threshold predominantly depends on the integrity of the IHC and not so much on the OHC system. Therefore audiometric evaluation is insufficient for determining the status of the OHC system along the basilar membrane and might, at best, be sufficient for approximate prediction of the presence of tinnitus. Independent evaluation of the status of IHC and OHC is needed. Methods developed recently for evaluating cochlear emissions and distortion products can be used for determining the status of

OHC^{17,84,131,133,134,163,168,210,211,279,285,299}, whereas traditional audiometry might be used to estimate the integrity of the IHC system.

Exposure to intense sound first results in bending of the OHC stereocilia, effectively decoupling them from tectorial membrane and preventing sound-induced excitation of OHC. If the noise is not too intense and does not cause permanent threshold shift, the stereocilia return to their normal state within hours or days. Transient tinnitus is frequently reported after exposure to such noise^{64,135,169}, which is consistent with the proposed theory. Permanent damage of OHC might result in permanent tinnitus, provided that there are still some functional IHC in the area. Interestingly, comparison of sound-induced tinnitus with chronic tinnitus indicated that both kinds may arise from the same auditory precesses⁶⁴.

2.4 Calcium involvement in cochlear dysfunctions

There is consensus that calcium plays a significant role in cochlear function^{2,49,57,59,99,101,104,227,252,268,297}. There are several possible mechanisms through which calcium might influence the transduction properties of the cochlea. Calcium has been shown to influence: (i) position of the tectorial membrane¹⁴⁰; (ii) hair cells' calcium-dependent potassium channels^{32,35,98,100,101,227,269}; (iii) transduction properties of hair cell cilia¹⁹¹; (iv) slow motile properties of OHC^{59,107,268,296,297}; and (v) release of neurotransmitter from hair cells^{37,252}. Additionally, hypothyroidism with associated low calcium is affiliated with hearing loss¹⁷⁵, presumably through development of a non-functional tectorial membrane as a result of low calcium level²¹⁶.

Previously, I have hypothesized that changes in calcium ion concentration in the perilymph as well as within hair cells may be responsible for a variety of cochlear dysfunctions including tinnitus, further implying that calcium channel modulators might be useful in alleviation of cochlear and vestibular disorders^{111,112}. This hypothesis is based on the following reasoning.

2.4.1 Effect on micromechanical coupling of the cochlear structures Decoupling of cilia from the tectorial membrane might result from decreased concentration of calcium in cochlear fluid, as this coupling is calcium dependent. Kronester-Frei¹⁴⁰ showed that removal of intracochlear calcium causes rapid swelling of the tectorial membrane, with a parallel increase in the distance between the hair cell cilia and the membrane. Further on, the ultrastructure of the tectorial membrane matrix loses its organized striated sheets, disintegrating into randomly dispersed fibrillar material and small particles⁸⁶. Since even minor changes in the parameters of ciliary coupling with both hair cells and the tectorial membrane have a significant impact on the transduction properties of hair cells, calcium-induced changes in coupling could be a physiological basis for Tonndorf's hypothesis.

In this regard, it is interesting that subcutaneous or oral administration of 300–400 mg/kg of aspirin or sodium salicylate in rats causes a significant decrease in serum calcium^{118,126,127,222}, to the extent sufficient to be responsible for fetal toxicity. Moreover, this teratogenic effect of salicylate can be prevented by exogenous CaCl_2 ²⁶⁷. Our observation fully confirmed these findings, further indicating that salicylate decreases calcium levels in cerebrospinal fluid (CSF) and perilymph^{118,120}. Interestingly, our experiments with another tinnitus-inducing drug, quinine, showed both behaviorally detectable tinnitus¹¹⁶ and a significant decrease of serum calcium (Jastreboff, in preparation).

2.4.2 Outer hair cells In this respect, the recent report of Brownell and co-workers, analyzing electrically and chemically induced OHC volume changes, is of particular interest²⁰. Brownell postulated that 'The low calcium may increase the permeability of the

cytoplasmic membrane to ions and low molecular weight molecules'. Therefore, decrease of calcium results in increase of OHC diameter and decrease of OHC length under normal conditions. This, in turn, might be responsible for the reported attenuation of cochlear active processes by salicylate due to mechanically decoupling OHC from the tectorial membrane, in addition to the direct effect of salicylate on the electrical-mechanical conversion site in the OHC membrane⁶.

Other possibilities of tinnitus generation are based on the involvement of calcium in transduction processes elsewhere within the cochlea. There is consensus that calcium is required in the process of transduction of mechanical stimuli into electrical potentials^{31,32,37,98,184,269}. It has been shown that acoustic transduction is mediated by calcium-sensitive potassium channels^{32,100,227,269}, and that calcium plays a crucial role in determining the mechanical properties of cilia¹⁹¹ and the slow motile properties of OHC^{294–297}. Decrease of intracellular calcium concentration within the apical part of OHC should decrease the tension of the contractive proteins attached to the cilia's rootlets, changing their mechanical characteristics particularly for small displacements of the cilia. Thus, there might be an increase of 'free play' in ciliary pivoting and a decrease of mechanical coupling between the tectorial and basilar membranes, resulting in an increase of thermal noise and tinnitus. Although calcium is not essential for fast motility of OHC^{5,228,295}, it is crucial for slow OHC movements^{49,59,268,297}, which have been postulated to be involved in adjusting the basilar membrane position to assure the most efficient sound transduction in a wide range of intensities^{147,148,296}. These hypotheses predict that decrease of the intracochlear calcium level might result in non-linear, recruitment-type characteristics of cochlear microphonics (CM), particularly when recorded from OHC or from outside of the cochlea, when OHC activity predominates in CM³⁷. Increase of noise in CM and cochlear action potentials (CAP) might be expected as well. Clinically, this type of tinnitus should be associated with worse speech perception than might be predicted from the audiogram, with display of disproportionately poor speech discrimination near threshold and better discrimination at higher intensities.

It is noteworthy that ionic compositions are not homogenous along the cochlea and concentrations of endolymphatic calcium have been reported to increase by a factor of two between the basal and apical parts of the cochlea¹⁰⁵. This finding might be partially explained by recent data showing that longitudinal flow of endolymph does not exist²²⁴. Thus local ionic concentrations represent a dynamic balance between passive diffusion and active pumping of the ions from stria vascularis^{104,124,190}, which can be different for different portions of the organ of Corti. The distribution of hair cell stereocilia and their interlinking is heterogenous as well, with taller cilia and higher extent of interlinking occurring in the basal part of cochlea^{63,257–259}. Given the information above, it is reasonable to postulate that a change in serum calcium level, which has been shown to be reflected in calcium concentrations in CSF and perilymph¹²⁵, might influence various parts of the basilar membrane to different extents, perhaps with the basal end of the basilar membrane affected first. Therefore, even a general change in the serum calcium level might result in tonal type of tinnitus.

2.4.3 Modulation of neurotransmitter release Another interesting possibility is the direct influence of calcium level on the release of neurotransmitter from hair cells. Data obtained from lateral line preparation showed that gradual increase of calcium level resulted initially in suppression of spontaneous activity of the VIII nerve fibers, while evoked activity tended to increase^{45,46,80,187,242}. Only when calcium level was further increased above 5 mM were both types of activity suppressed. Conversely, a decrease in

the calcium level increased spontaneous activity and decreased evoked activity. For low calcium levels, evoked activity was totally abolished while spontaneous activity remained stable, but with a change from random to regular type of discharges⁴⁵. These data indicate the possibility of the existence of partially independent mechanisms of transmitter release for spontaneous and evoked activity, both of which are calcium-dependent, but in a different manner. This postulate is further supported by the finding that venom from the spider, *Argiope trifasciata*, a highly specific blocker of ion channels associated with glutamatergic receptors, in lower doses affects sound-evoked activity recorded from auditory nerve fiber without influencing spontaneous activity³⁶.

Although there is consensus that calcium is essential for transmitter release, the novelty is that modification of the calcium level might have a converse action on spontaneous versus evoked activity. Since a decrease in the calcium level results in an increase of spontaneous and a decrease of evoked activity, this should cause a decreased signal-to-noise ratio, with increased tendency to tinnitus and the appearance of a problem with speech perception and hearing in general.

In summary, it is possible to predict the following sequence of events when the calcium level decreases in the cochlea. The tectorial membrane will swell, increasing the distance between the membrane and OHC¹⁴⁰, causing partial mechanical decoupling, which should increase the thermal noise of the system^{85,260}. Transduction characteristics of the cilia will change in a manner indicating a more flexible attachment¹⁹¹, which again should decrease the mechanical coupling between OHC and its cilia, resulting in decreased mechanical coupling between the tectorial membrane and OHC with the consequences as described above. Slow mobility of OHC will decrease^{191,296}, reducing the influence of feedback control extended by the efferent system on OHC-controlled micromechanical properties of the basilar membrane, which in turn might prevent active compensation for local damage of hair cell systems. The OHC membrane permeability to ions and low molecular weight molecules will increase, resulting in changes of OHC volume and turgor²⁰. Spontaneous activity of the VIII nerve will increase while auditory evoked activity will decrease^{45,46,242}. All those effects are proceeding in one direction – increase of the noise within the system and parallel decrease of signal-evoked activity. This will result in deterioration of the signal-to-noise ratio, which would facilitate the appearance of tinnitus. Additionally, this should reduce recognition of sounds, particularly in noise and as complex as speech.

The calcium level influence might be of interest from a clinical perspective, since abnormal increase of spontaneous activity and its increased regularity might be responsible for some types of tinnitus. Therefore by increasing the cochlear calcium level, it should be possible to suppress or eliminate tinnitus, with the further possibility of enhancing evoked activity. Thus, it might be possible to alleviate tinnitus without compromising hearing.

2.5 Calcium channels

Transmitter release, as well as slow motile properties of OHC, depend on intracellular calcium level^{37,79,139}. Although extracellular calcium will influence intracellular calcium concentration, the crucial role is played by functional characteristics of the calcium channels¹⁷⁶. Since calcium influx is an intermediate step in transmitter release from hair cells, and since certain types of tinnitus might result from increased spontaneous activity, it is possible to expect that by inhibiting calcium influx into hair cells, it should be possible to attenuate some cases of tinnitus originating in the cochlea.

Another possibility relates to the postulate that calcium-dependent slow movement of

OHC might be involved in adjusting basilar membrane position^{147,148,295,296}. Therefore by affecting the function of OHC calcium channels, it is possible to expect changes of the basilar membrane position as a result of calcium channel modulator action.

This hypothesis has found preliminary confirmation in our data, where it was shown at the behavioral level that a calcium channel blocker, nimodipine (BAY e 9736), abolished salicylate-induced tinnitus in animals^{111,112,116}. Nimodipine acts on L-type channels, prolonging the inactive state of the channel⁹⁶. This substance has the interesting ability to pass through the blood-brain barrier much better than any other known calcium channel antagonist^{129,239}. Possible applications for this drug focus on the described increase of the brain blood flow by nimodipine, thus suggesting its use in cases of subarachnoid hemorrhage^{3,25,239} and ischemia^{128,250,251}.

Recently accumulated data indicate the possibility of nimodipine's interaction with CNS in a direct manner²³⁹. Our data, recording the characteristics of cochlear potentials from the intact cochlea^{111,112} and from cochleas perfused with artificial perilymph containing different doses of nimodipine^{12,13}, clearly demonstrated that nimodipine exerts its action directly on the cochlea and hair cells without changing cochlear blood flow. Additionally, our data indicate that the action of the drug occurs at least at two levels, perhaps affecting slow motile processes within the OHC system and has an impact on the processes involved in transmitter release. Furthermore, another calcium channel blocker, MK-600, while in large doses has been shown to inhibit both spontaneous and evoked activity, in lower concentrations this substance suppressed spontaneous activity without affecting evoked activity⁸⁰. Clinical data, although preliminary at this moment, support the possibility of suppression of tinnitus by nimodipine²⁵⁵.

2.6 *Aging, calcium and L-type calcium channels*

The hypothesis relating calcium with dysfunctions of the auditory and vestibular systems has several implications. Since many of the mechanisms discussed for cochlear hair cells should exist for vestibular hair cells, one might expect similarity in mechanisms underlying normal function and malfunction between both systems. For instance, it is possible that disorders resulting in aberrant calcium levels in CSF and perilymph might increase susceptibility of both the auditory and vestibular systems to malfunction. In addition, it might explain the tendency of the elderly toward tinnitus and vestibular disorders. This postulate is based on the following observations.

It has been proposed that altered calcium homeostasis is involved in the process of aging, with modified calcium permeation across membranes a common change^{21,42,70,123,143-146,206}. Available data imply that general calcium metabolism is disturbed in the elderly¹⁴⁹.

Salicylates, well known for increasing the threshold of hearing and inducing tinnitus in humans^{61,169}, cause decreased calcium levels in serum and CSF^{118,120,126,127,222,267}. A report published recently on salicylate toxicity in people of various ages showed that the elderly are more susceptible to the effects of this drug, and that smaller doses per kg of body weight are sufficient to induce tinnitus⁷⁶, in spite of essentially unchanged salicylate uptake and plasma clearance in humans¹⁸³. We reported that calcium levels decreased in serum, CSF and perilymph after salicylate^{118,120}. Our data show that this decrease is highly non-linear as a function of dose and salicylate serum levels¹¹⁸. The threshold level of salicylate-induced calcium decrease corresponds to the reported level that evokes tinnitus in humans⁶¹. The threshold of calcium decrease is related to the animal's age, with a lower value observed in older animals (Jastreboff, in preparation).

Auditory system dysfunction in the elderly and the high frequency of tinnitus occur-

rence, affecting about 1/3 of the population above 65 years of age, has a profound effect on their life^{81,223,231}. Although there are reports describing anatomical and physiological changes in animals' hearing^{23,102,142,280–283}, nevertheless, no theory has been presented as yet to explain why the elderly have such a high incidence of tinnitus and vestibular disorders, except to associate it with age-related noise or drug-induced cumulative hearing loss.

I postulate that disturbances in calcium metabolism and dysfunction of L-type calcium channels reported for the elderly and aging tissues^{4,21,44,106,143–146,206,291} might affect cochlear function, increasing the spontaneous activity of auditory nerve fibers among other things. This might enhance changes in the auditory system due to other minor pathologies which would otherwise be subthreshold in terms of tinnitus perception. Additionally, it might enhance the severity of tinnitus already existing due to other mechanisms. The observation describing a correlation of senile hearing loss to decreased serum calcium level and disturbed calcium metabolism further support this postulate²⁹¹.

A similar effect could exist in the vestibular system where calcium ions play an important role^{32–34,50,101,227,252,269}, and their imbalance has been implied as being responsible for Meniere's disease and other vestibular disorders^{172–174,188}. Alternatively, it has been proposed that Meniere's disease and associated tinnitus results from potassium poisoning of cochlear hair cells, due to the flow of potassium-rich endolymph to perilymph through rupture of the scala media caused by endolymphatic hydrops^{138,236,293}. Nevertheless, this does not exclude the facilitatory influence of calcium imbalance, with potassium poisoning being the final step.

The loss of calcium homeostasis could also facilitate the increase of vestibular disorders seen in the elderly. Interestingly, melanocytes have been reported to act as a calcium buffer^{172,173,185}, and the relationship between pigmentation and susceptibility to auditory and vestibular disorders (higher pigmentation – higher resistance)^{28,29} further support the possibility of calcium involvement. Relevance of melanin for pharmacological treatment of tinnitus has been proposed as well¹⁶⁴.

Continuing this theme, calcium channel modulators should play an important role in the function of the vestibular and auditory systems and might be useful in alleviating their disorders. The process of aging has been related to increased calcium influx into cells¹²³. The calcium L-type channel antagonist, nimodipine, was reported to be effective in improving the performance of old animals in behavioral tasks⁴², in hippocampal long-term potentiation^{21,145}, and in preventing deterioration of the vestibular and motor systems^{73,237–239,270}. Nimodipine crosses the blood–brain barrier better than any other calcium channel antagonist, therefore therapeutic levels in CSF can be reached without toxic side-effects^{129,239}. Our data support this therapeutic possibility^{111,112}.

Thus, it is proposed that calcium channel modulators might be effective in preventing age-related deterioration of the auditory system in a manner similar to that described for vestibular and motor systems^{237,239}. Although more work is needed to prove this hypothesis, it provides a testable hypothesis for the mechanism of aging of the vestibular and auditory systems. Additionally, it proposes a new approach for treatment of auditory and vestibular disorders, indicating a possible etiology for disorders associated with aging in these systems and a strategy to alleviate or slow down some age-related processes.

2.7 Synaptic transmission

Another cause of cochlear tinnitus can be related to increased synaptic transmission between the IHC system and the VIII nerve. This might result from several factors, such as: (i) increased spontaneous release of a transmitter substance from the IHC, thus

increased spontaneous activity within the auditory nerve; (ii) increased sound-evoked release of this transmitter; (iii) increased effectiveness of the transmitter on the postsynaptic membrane; (iv) decreased inhibition caused by the efferent system.

The first two possibilities are related to a dysfunction within the hair cells. It is possible to envision several mechanisms which might, in turn, enhance spontaneous release of the transmitter. Perhaps the simplest mechanism involves the effect of modulation of calcium concentration as discussed previously. Modification of calcium concentration within the hair cells depends on its concentration in the perilymph¹²⁴ and in the kinetic properties of calcium channels. Alteration in calcium concentration within the perilymph indicates only the possibility of changes within the hair cells, while modification of the calcium channel should have a direct impact on the intracellular calcium level. Thus, one may speculate that increased calcium influx into the hair cells through dysfunctioning calcium channels should result in increased spontaneous release of transmitter, causing increased spontaneous activity within the VIII nerve. The observation that calcium decrease has an opposite effect on spontaneous and evoked activity^{45,46,242} suggests the possibility of separate mechanisms governing evoked versus spontaneous transmitter release.

Modification in transduction parameters resulting in increased transmitter release from the same mechanical stimuli (displacement of the cilia) might be related to higher sensitivity of calcium-controlled potassium channels involved in the first stage of transduction, or in increased sensitivity of calcium channels involved in the second stage of transduction.

The third possibility involves increased sensitivity of the postsynaptic site. Tonic partial depolarization of the postsynaptic membrane, or decreased activity of a catabolic enzyme digesting a neurotransmitter, which will result in accumulation of the neurotransmitter and its prolonged action on the receptor, might produce such a effect. This can be tested and discussed in more detail once the afferent neurotransmitter and its catabolic enzymes in the cochlea are positively determined^{11,10,11,47,58,79,122,139,187,214,242}. However, it might be predicted that dysfunction of this kind should be reflected in elevated characteristics of CAP as compared to CM, particularly for close-to-threshold stimuli. A faster saturation of CAP as compared to the control situation can be envisaged. At the clinical level, patients with decreased loudness discomfort level might belong to this category, together with patients with decreased efferent inhibition, which is described below.

The fourth category concerns the decreased efferent inhibition of the afferent fibers coming from IHC. Again, this should result in relatively larger afferent excitation after the same amount of neurotransmitter is released. Possible effects resulting from dysfunction of the interaction of efferent fibers with the OHC system are discussed separately. It should be stressed that in addition to general disinhibition involving all IHC or OHC, tinnitus might result from even relatively small changes occurring only in a limited (frequency-specific) portion of the efferent system, which might be further enhanced by the 'edge effect', as discussed below.

2.8 Edge effect

In the visual system, the enhancement of contrast in the transition between patterns of different density has been documented²⁴⁵. There is consensus that the major factor responsible for this phenomenon is lateral inhibition, and that the edge effect exists for modalities other than vision⁷⁸. It has been postulated that a decrease of activity in only a subpopulation of neurons, further enhanced by lateral inhibition, could be responsible for tinnitus^{137,226}. Observation of local decrease in the activity of some auditory nerve fibers

after cochlear damage was used to support this postulate. Furthermore, it has been indicated that tinnitus patients frequently report the pitch of tinnitus as corresponding roughly to the transition point from normal hearing to that of an adjacent frequency of elevated threshold.

This hypothesis can be extended further by saying that any localized change in cochlear function or VIII nerve activity might result in tinnitus, since a lateral inhibition-based edge effect will act on both a decrease and an increase in activity. This local change might result from mechanical damage, e.g. noise-induced or age-related effects. The point is that due to the edge effect, either an increase or decrease of local neuronal activity, presenting even small variations along the cochlea's partition, may have a great impact after being enhanced by edge effect.

This should be true for both modified spontaneous or evoked activity, synchronization of that activity, or abnormal temporal patterns of discharges, as long as the change is localized. In other words, if the activity of abnormal hair cells, VIII nerve fibers, or cluster of neurons within a nucleus shows an abnormal pattern of activity with a clear border from normally functioning units, one might expect that this difference will be further enhanced by an edge effect. Although physical closeness is easiest to imagine, it can be postulated that functional interrelation might act in the same way. If cells which are even physically distant but which project to the same part of a higher center (the same group of neurons), or are part of a neuronal network performing a specific function, exhibit an abnormal type of activity, then edge effects could occur. This will be further discussed in Section 3.

2.9 Efferent system

Efferent innervation exerts a clear effect on the OHC system^{152,247,276}, IHC afferents^{71,72,277,278,289}, otoacoustic emission^{186,189}, and on cochlear evoked potentials¹⁷¹. Hazell suggested a mechanism of tinnitus generation which involves control of active processes in the cochlea by the efferent system⁸⁸. The cochlear efferent system affects both classes of afferent fibers coming from IHC and OHC. The OHC play a crucial role in sharpening the tuning of the basilar membrane through active mechanical processes and results in negative dampening of the basilar membrane¹⁴⁷. The possibility that an overreaction of this active system might yield tinnitus has already been proposed^{87,88,132,200,286-288}.

Motile activity may occur in a cycle-by-cycle manner controlled by stress-activated ion channels acting in phase with basilar membrane displacement. The efferent system may act in a tonic manner, adjusting the working point (e.g. the initial tension exerted by the OHC) as proposed by LePage¹⁴⁷. The efferent system is, in turn, under the control of feedback from auditory information provided by the afferents exhibiting a high degree of frequency selectivity, frequently having equally low threshold of response and equally sharp tuning as afferent fibers¹⁸. When damage of a group of outer and/or inner hair cells occurs, there is decreased auditory information from this region. This might result in a localized reduced efferent discharge rate. Since efferent innervation is diffuse^{18,19,151,153}, the ensuing disinhibition would affect normal undamaged hair cells adjacent to the damaged area, resulting in increase of negative overdampening⁸⁸. These, in turn, will yield enhancement of the ambient or thermal noise of the system at a frequency corresponding to the position of the border of the damaged area on the basilar membrane. The effect may be further enhanced by higher centers, the overall result being the perception of tinnitus.

At the experimental level, it should be possible to detect increased noise in both the

CM and CAP, as well as non-linearity of the CM and psychological perception for tones very close to the frequency(ies) of tinnitus. To detect this phenomenon, it would be necessary to measure these potentials for close-to-threshold perception in very small frequency increments. An increased response and a decreased threshold would indicate the presence of a section of basilar membrane with negative overdamping. Interestingly, decreased threshold of hearing for frequencies close to tinnitus has been reported²⁶⁶.

The clinical prediction is that this type of tinnitus would be very difficult to mask using external auditory signals because negative overdamping would result in decreased discomfort levels and increased perception of the external signal with a pitch close to that of tinnitus. Total auditory masking of tinnitus, defined as total suppression of tinnitus perception by external sound, should occur on the basis of two-tone inhibition, therefore for tones being some distance from the perceived pitch of tinnitus. Masking frequencies close to the tinnitus pitch should be effective only at relatively higher levels, if at all.

2.10 Cross-talk between hair cells or VIII nerve fibers

Most views on the origin of tinnitus are based on the assumption that the location of active neurons, as well as the mean activity of their discharges, are the principal determinants of auditory processing. Alternatively, Møller¹⁷⁷ postulated that tinnitus is based on the occurrence of 'cross-talk' between demyelinated auditory nerve fibers or hair cells where there was breakdown of electrical insulation between them. Such a loss of insulation could occur following acoustic neuroma, vascular compression syndrome, and other retrocochlear pathologies^{108,109,178-181}. This could result in increased phase correlation of the spontaneous activity of different fibers, which ultimately become phase-locked with each other, leading to abnormal pulse generation.

2.11 Analogy with pain

Recently, Tonndorf proposed a mechanism of tinnitus based on an analogy with pain perception²⁶². According to this theory, tinnitus results from an imbalance between the activity of large fibers innervating IHC and small OHC fibers, caused by partial damage of one or both of the systems. This model was further extended to imitate Melzack's Gate Control Theory of pain, arguing the similarity of IHC fibers to somatosensory, large-diameter fibers and OHC fibers to small-diameter, pain-related fibers. It remains to be shown that the interaction mediated by the fibers' innervation of OHC and IHC occurs in the manner of presynaptic inhibition, as required in Melzack's and this model.

The relationship of changes between cochlear dysfunction and tinnitus outlined above represents only an example of possible scenarios. Practically any local modification of cochlear transduction properties might produce tinnitus due to a contrast enhancement between altered and normal areas which might be amplified by further central processing within the nervous system, as discussed below.

2.12 Hyperactivity within the auditory pathways

It is generally agreed that the CNS compensates for a decreased information input in any modality by increasing the sensitivity of centers involved in perception. The action of the CNS is oriented to homeostasis, with excitatory and inhibitory influences interacting at all levels, thus increasing its flexibility and control and providing fine balance, as well as adaptation and habituation. Total absence of input disturbs such a balance and might result in abnormal activity of centers involved in the processing of this information. I am proposing a hypothesis that even for tinnitus originating in the cochlea, processing of tinnitus-related information within the auditory pathways is of crucial significance, rather than assigning to those pathways a role that is one of purely passive transmission.

Gerken and co-workers reported that after sound-induced hearing loss, sensitivity of the cochlear nuclei and inferior colliculi neurons to electrical stimulation increased and recruitment emerged^{14,66-69}. This observation was strengthened further by Sasaki et al.^{229,230}, who reported a relative increase of 2-deoxy glucose uptake into cochlear nuclei and inferior colliculi after damage of the cochlea and auditory nerve starting a few days after the surgery. Salvi and Ahroon²²⁶ presented results similar to those of Gerken and co-workers, while increased neural excitability in inferior colliculus related to noise-induced hearing loss has been reported by Willott and Lu²⁸⁴.

The mechanism of this phenomenon is unknown. One may speculate that decreased input from the periphery causes contraction from the neuronal assembly involved in automatic gain control within the auditory nervous system causes plastic, long-lasting changes in synaptic weight. These finally result in increased sensitivity of those nuclei to any input, including spontaneous activity which, in turn, might yield the perception of tinnitus. Another explanation is presented in Section 3.

It is possible to induce tinnitus in some people by attenuation of an external acoustic signal and to decrease it by enhancing the auditory input (e.g. by hearing aids)^{94,169}. This might be partially explained by peripheral auditory masking provided by amplified environmental sounds, but the fact remains that for many patients this effect is not instantaneous and might take days⁹³, indicating the need to consider plastic changes in the CNS. From this point of view, the surprising and unexplained clinical phenomenon is that of total deafness with no tinnitus whatsoever.

All tinnitus, whether generated peripherally or centrally, finally results in abnormal cortical activity, as this is where it is perceived. However, cortical tinnitus occurs in the absence of an abnormal subcortical input. At the present moment, the only certain model of tinnitus other than the cochlear type is where both acoustic nerves have been sectioned. This is obviously not a proper model of cortical tinnitus, since such an operation will result in increased subcortical sensitivity, as discussed previously.

Theoretically, it is possible to envision significant differences between the hyperactivity occurring at the cortical level as compared with that in the more peripheral pathways. One might expect very complex, hallucinatory-like perceptions of tinnitus, which may be then categorized as auditory hallucinations. Involvement of other cortical areas becomes more likely (hypersensitivity to light or tactile stimuli may be evident)⁹⁰.

Summarizing possible mechanisms of tinnitus generation, it is hypothesized that changes in cochlear micromechanics, and more generally in dysfunctions within the cochlea, could produce tinnitus through several intermediate steps. Of particular importance might be extra- and intracellular calcium levels. Dysfunctions of the efferent system and active processes occurring in the cochlea could be another important cause of tinnitus. Tinnitus might result from abnormal hyperactivity of the higher levels of the auditory pathways, including the auditory cortex. I realize that it is possible to propose a wide variety of detailed hypotheses based on micromechanical, biochemical or even molecular abnormalities, but rather the intent was to outline classes of possible tinnitus generators.

3. DETECTION OF TINNITUS-RELATED ACTIVITY

3.1 *Unusual psychoacoustical features of tinnitus*

Perception of tinnitus exhibits a number of features which are quite different from the perception of external sound of similar intensity and frequency spectrum. Contrary to masking of external sounds, tinnitus can be masked by pure tones or band-pass noise of a

wide range of frequencies, often contralaterally with the same effectiveness and with intensity of masker, frequently abnormally great or small as compared to the signal-to-noise ratios required by external signals and maskers^{55,94,169,192,195,197,198,203,204,265,272}. Indeed, masking of tinnitus resembles central masking of external sounds. Contrary to masking external sounds, tinnitus requires increased intensity of masker over a period of minutes^{193,197,198,265}, while external sound does not require this increase. Sometimes it is impossible to mask tinnitus, disregarding intensity of external sound²⁷³. Furthermore, it is impossible to create beats by interference of tinnitus with external tone, and tinnitus does not exhibit phase-related phenomena¹⁶⁹. Cessation of masker is frequently followed by a period of residual inhibition when tinnitus disappears or is reduced in magnitude^{27,169,199,203,273}. Additionally, tinnitus does not habituate even when its perceived loudness is weak.

3.2 *Abnormality of tinnitus-related patterns*

These observations suggest that the representation of tinnitus within the auditory pathway may be different from the representation of external sounds. I hypothesize that, in the majority of cases, tinnitus results from the perception of abnormal activity within the auditory pathways which cannot be evoked by any combination of external sounds. Additionally, I propose that detection of the tinnitus signal occurs in a pattern-detection manner by neuronal assemblies, and that this detection process undergoes plastic changes. In the remaining part of this section, I will attempt to show how those two postulates can untangle otherwise unexplained puzzles of tinnitus.

The simplest situation with abnormal activity occurs in the condition when neighboring fibers fire in synchrony¹⁷⁷. Under normal conditions, it is possible to partially synchronize a group of fibers by low-frequency tones (each fiber will fire with higher probability within a given phase range), but the phase will be different for each fiber. Even then, this synchronization will not include physically close fibers, since even fibers innervating the same IHC exhibit different thresholds of activation and different phase relations in reference to an external tone. Since tonotopic organization is preserved at each level of the auditory system, physically close fibers will also be close in characteristic frequency and should project in a like manner to higher-order neurons. Synchronized, even weak activity will evoke much higher excitation of the goal cells than stronger but diffused tone-induced activity. Such synchronized activity might be much more difficult to attenuate because of its unusual pattern. Any other abnormal temporal and/or spatial patterns of activity within the auditory system might induce disproportionately strong excitation of higher-order neurons. Lateral inhibition and pattern-enhancing features of neuronal assemblies could produce further enhancement.

Although there is only very limited information available on neuronal activity which might be related to tinnitus, nevertheless these data are in agreement with the proposed hypothesis of abnormal patterns of discharges^{53,54,121,226}. Moreover, an analysis of tinnitus-related abnormal patterns of activity should provide interesting information for work on auditory prostheses, when electrical stimulation of the auditory nerve (cochlear implants) or cochlear nuclei complex, which results in abnormal patterns of excitation, is used to induce perception of sound in a deaf ear.

3.3 *Deciphering tinnitus puzzles*

Initially, a totally new tinnitus-related signal has to be classified and categorized. This stage might follow the process of self-organization of recognition categories²² in a manner described for a self-organizing recognition system able to build codes for speech signals, as proposed by Grossberg⁷⁸. This theory incorporates adaptivity of the networks involved,

development of codes for new input patterns with stress on the necessity of attentional and orienting subsystems to process familiar and unfamiliar events, code stabilization with its involvement in suppression of noise in input patterns, postulate rescaling of noise criterion to each pattern context, and stresses the importance of two-way interaction of the subsystems involved^{22,78}. The detailed description of this interesting mathematical theory is beyond the scope of this paper. Only implications for tinnitus recognition and perception theory are incorporated in this text.

3.3.1 Detection After crossing the threshold of pattern detection, tinnitus-related weak signals, immersed in the noise of spontaneous activity, are processed by neural assemblies. The initial newness and persistence of the signal, combined with the alarm that tinnitus usually generates, enhances the recognition of this particular pattern and induces plastic modification of adaptive filters. This assures that future appearance of this particular pattern, even if contaminated by noise, will be recognized. The process requires some period of time and repetition of the initial signal, but unfortunately for patients, tinnitus is persistent and the initial anxiety induced by its presence very strong, reinforcing tuning of the system to the tinnitus signal and effectively improving detection of tinnitus. *

The initial process of recognition involves crossing a detection threshold of a relatively weak tinnitus-related signal and separating it from spontaneous activity. The threshold-crossing phenomenon might contribute to the very rapid onset of tinnitus reported by patients. Typically, patients describe the onset of tinnitus as a rapid phenomenon, appearing in full strength from the very beginning. Due to the abnormal pattern of this activity, it might be much more difficult to suppress (or habituate to) than to external sounds. Furthermore, if tinnitus detection is carried on by neuronal assemblies in a manner related to neuronal networks or a holographic neuronal network^{22,62,78,160,208,209,240,275}, even incomplete or weaker patterns similar to the original one will evoke nearly the same perception.

This hypothesis might explain the persistence of tinnitus perception even when a peripheral generator produces a transitory, fluctuating or intermittent signal. The remaining weaker signal is still sufficient to induce detection in a network which has been 'tuned' to the tinnitus pattern by creating a recognition category for tinnitus-related activity^{22,78}.

Moreover, if the tinnitus-related pattern, as I postulate, cannot be imitated by any combination of external sounds, its masking, attenuation and habituation should be quite different from that of external applied sounds. As has been pointed out, the simplest example of such an abnormal pattern will result from electrical stimulation of the cochlea. Reports of patients with unilateral deafness who have undergone electrical stimulation of the deaf ear support the proposed hypothesis of abnormality of tinnitus-related activity.

* This process can be illustrated by the observation of a person exposed to a totally new language. Initially, it is impossible to detect any words or even familiar sounds, and the language sounds like a strange noise or melody. After the initial learning stage has been accomplished, suddenly it is possible to divide phrases into fragments and recognize familiar repetitive patterns, even without the slightest understanding of the meaning of the phrases. This stage represents the transition from treating new sound patterns as a non-specific noise, to pattern recognition of crucial features of the language. Once this is achieved, it is usually never lost. A new pattern of excitation within the nervous system, if weak, is treated as part of the noise of spontaneous activity and ignored. When the signal is stronger or more persistent (even though still weak), it is detected as a new significant pattern, and attenuation of its perception becomes difficult, e.g. it requires strong masking by an external noise source.

The same problems encountered in reconstructing the sound of tinnitus with the use of a music synthesizer appeared when trying to imitate electrical stimulation-evoked perception by providing the hearing ear with a wide variety of synthesized sounds. Even for simple, sinusoidal electrical stimulation, perceived sound is complex and practically impossible to imitate⁹⁰. The psychoacoustical data obtained in evaluation of patients with cochlear implants show that even with most advanced multichannel implants, and with elaborated sound processors, the quality of the perceived sound is poor, with a small dynamic range and very limited frequency and intensity discriminations^{30,92,207}. In only 5% of the cases is speech recognition without lip reading possible, and an appreciable number of patients with implants (2–15%) choose to discontinue the use of their prostheses³⁰. These observations are in agreement with data from single fiber recordings from the auditory nerve, with electrical stimulation of the cochlea showing abnormal histograms of activity and compressed, steep rate-intensity functions^{14,207,246}. Furthermore, when contralateral acoustical masking of electrically evoked sensation was evaluated, typical flat characteristics of tinnitus masking were obtained⁹⁰.

I would like to stress that as tinnitus results from a variety of causes, there is a continuum of tinnitus-related activity from such causes, which can be closely imitated by external sounds to activity which cannot be induced by any combination of external sounds. Depending upon the extent of abnormality, it should be easier or more difficult to mask or otherwise lessen tinnitus. I postulate that tinnitus resulting from only slightly abnormal activity will have undergone habituation, as external signals do. Therefore in practice, cases are seen of tinnitus resulting from significantly abnormal activity.

3.3.2 Masking One of the puzzles of tinnitus is the observation that in different patients reporting tinnitus of the same loudness and pitch, sometimes it was necessary to use much higher levels of external noise to mask the tinnitus than would be expected from the psychoacoustical masking level of a comparable external tone: in others, for practically the same sensation of tinnitus, a very low level of external noise is sufficient to suppress the perception of tinnitus completely^{55,94,193,197–199,203,264,265,273}. This phenomenon can be explained by proposing that, in the first case, tinnitus results from neuronal activity which is highly different from any activity induced by sound, while in the second case, tinnitus-related activity is close to patterns of activity induced by sound. Therefore, it is possible to retune the neuronal assembly detecting tinnitus by increasing the system noise, thus retuning the neural assembly away from a tinnitus pattern. This phenomenon cannot be explained by peripheral masking of tinnitus. The problems involved with masking will be discussed in detail elsewhere (Jastreboff and Hazell, in preparation).

3.3.3 Residual inhibition Recognition of patterns of variable complexity in noise²² and plasticity scheme may also explain a phenomenon known as ‘residual inhibition’^{26,27,55,94,198,199,203,272,273}, e.g. total disappearance of tinnitus perception (commonly for seconds, minutes, hours, or occasionally days) after masking by external sounds, yet another mystery of tinnitus. When tinnitus results from relatively normal neuronal activity, detection of even relatively loud tinnitus may occur at the border of the ‘detection space’ in a metastable, adaptive network. Hence when the equilibrium of the network is disturbed by increasing the level of additional external noise (the prerequisite for residual inhibition to occur), the system switches to another metastable state in which the same tinnitus signal is below the threshold of detection. After switching off external noise, this metastate can be sustained for some period of time, up to the moment when the system switches back to its previous preferred state, with resultant tinnitus detection. Residual inhibition is the time needed to restore equilibrium. Repetitive masking might

cause gradual shifting in the tuning of the neuronal assembly, resulting in prolonged periods of residual inhibition, in some cases suppressing the detection of tinnitus totally.

3.3.4 Distributed involvement of entire auditory system I would like to stress that the process of recognition and classification of an activity within the auditory pathways might play a very significant role in the process of tinnitus generation and in the reported increased excitability of auditory pathways in a situation of abnormally low input from the periphery^{14,66–69,161,229,230,284}. The same principles of adaptive filtering and of distinguishing signal from noise by neuronal networks²² should apply. An imbalance in the activity, intensified by lateral inhibition and contrast-enhancement schemes, might result in abnormally amplified activity in the auditory pathways. Dysfunction of neuronal networks involved in pattern recognition and classification, including, for example, abnormally high feedback from attentional and orienting subsystems²², might aberrantly enhance an otherwise ignored anomaly in neuronal activity, yielding tinnitus perception. This points out that classification of tinnitus as peripheral, cochlear and central is superficial and is not advisable, since even if the initial cause of tinnitus can be traced to the cochlea, neuronal processing can be dominant. Therefore, the same cochlear damage in different humans might or might not produce tinnitus.

One implication from this model is that although tinnitus may appear very rapidly, its disappearance (at least for periods in excess of a few hours or days) is much more difficult to achieve and may require longer periods of time for treatment to be effective. Another aspect, discussed in the next section, is the importance of emotional factors and attitudes toward tinnitus for final tinnitus evaluation.

4. PERCEPTION OF TINNITUS

Up to this point, the possible mechanisms of tinnitus generation and the detection of tinnitus-related abnormal patterns of activity within the auditory pathways have been discussed. From both theoretical and clinical points of view, the next step – tinnitus perception – is of importance. Tinnitus has been described as generated in the cochlea, undergoing pattern recognition within the auditory pathway, and becoming reinforced by processing within subcortical centers. The next step is perception, psychological evaluation, and classification of this pattern.

4.1 Previous experience and emotional state

Presumably, associative cortical areas are involved in this process, relating the perceived pattern to previous experience and to biological significance of a signal. The interpretation of an auditory pattern has a profound impact on the reaction it invokes and on the threshold of its detection and recognition during repetitive presentation. The cry of a baby can be perceived as insignificant and easily ignored by a stranger, while for the baby's mother, this auditory signal can be very powerful and wake her from sleep. In this example, contrasting this effect to the same sound depends on the relation of a sound to the history of its evaluation.

Recognition strategies involve different processing of various signals resulting in their preferential recognition. The cortical evaluation of a signal involves its association with patterns stored in auditory memory. Any such pattern can be and often is related to some emotional state. Tinnitus may recall the sound of traffic, aircraft or central heating – most patients actually start with a search for a source of real sound⁹⁰. Signals perceived within any sensory system can be related to signals in different systems. Perception of those signals which evoke strong emotional states (positive or negative), as being of

particular importance and increasing attention to this particular signal, will be preferentially enhanced²².

Depending on associations with an emotional state, the same pattern of neuronal activity initiating tinnitus at the periphery might be perceived quite differently in various individuals, or in the same person in different situations^{90,91}. * This brings to mind the importance of evaluating tinnitus in relation to other patterns in memory and their emotional associations. Initially, tinnitus is predominantly associated by sufferers with something unpleasant, dangerous, or anxiety-provoking^{90,169}. This negative image of tinnitus strongly enhances the attention placed on it which, in turn, enhances the detection and perception of the particular tinnitus-related pattern. Through positive feedback involving short- and long-term memory^{22,78}, the pattern recognition mechanism is further intensified, making discrimination of this particular pattern from spontaneous and external sound-evoked activities all the easier. Thus even weak, only partially preserved patterns can be recognized as tinnitus due to previous network training.

Intuitively, it appears that the louder the perceived tinnitus is, the higher the level of annoyance it should induce. In reality, the situation is more complex. First of all, there is the problem of how to evaluate tinnitus loudness. Fowler introduced the comparison of tinnitus to contralateral sound of frequency matching pitch of tinnitus and concluded that in the majority of cases, tinnitus has very low loudness, below 10 dB sensation level^{27,94,196,266}. Others argue that this approach is misleading and that, rather, scales corresponding to subjective loudness perception, taking into account the recruitment phenomenon, should be used^{194-196,218,264,266}. Even with this approach, correlation of annoyance and loudness exhibits abnormalities²⁶⁶. The patient's data show rather large within-session and between-session variability and indicate that factors other than loudness contribute to perceived tinnitus annoyance. Therefore, association of the perceived sound of tinnitus with a patient's history and emotional state might play an important role. From this perspective, the possible involvement of the prefrontal cortex is of importance.

4.2 Prefrontal cortex

The prefrontal cortex has been proposed as being unique among cortical areas in its relationship with interoceptive, as well as exteroceptive, sensory domains and, therefore, the foremost structure that could synthesize the inner and outer sensory worlds⁷⁴. Studies performed since the middle of the eighteenth century have established an anatomical basis for common beliefs: that the prefrontal cortex stands at the 'common endpoint' for diverse afferent channels and is 'privy' to all incoming information^{74,75}. This attribute of the prefrontal cortex makes it a candidate for the integration of sensory and emotional aspects of tinnitus postulated above.

* A clinical example is seen in one patient with a pronounced social conscience for whom the hissing sound of tinnitus was in itself not particularly loud or troublesome, but evoked very strong images of prisoners being tortured with continuous noise⁹⁰. Another patient with a very similar type of tinnitus found it particularly intrusive during the day, affecting concentration markedly. Paradoxically, although before the appearance of tinnitus this patient had had a problem with sleeping, at night his sleep actually improved, as it evoked memories of a happy carefree childhood and of the water cistern which had often produced a loud hissing noise in his attic bedroom as he drifted off to sleep!⁹¹

This last example is of particular interest because it indicates that the same tinnitus pattern can induce different associations in the same person when combined with other stimuli, resulting in an opposite effect of tinnitus on the patient's life. Additional complementary signals (presumably darkness, the time of day, and other factors associated with going to sleep), together with the patient's previous experience, were necessary to change the evaluation and assignment of the same perceived signal.

In the 1950s, a lobotomy operation was performed on 20 patients exclusively because of tinnitus⁹. One might expect that disconnecting the prefrontal cortex should result in disassociation of sensory and emotional aspects of tinnitus, and similarly to its effect on pain perception, and should cause a decreased level of annoyance produced by tinnitus. The data not only confirmed this obvious prediction (all 19 patients who survived reported decreased annoyance of tinnitus) but out of those, 8 patients reported a decrease in the perceived loudness of tinnitus as well. The remaining 11 patients reported tinnitus to be the same but that it was not bothering them to the previous extent, and none reported increase in tinnitus loudness.

This result shows that disconnecting the prefrontal cortex from the system evaluating tinnitus actually interfered with the process of psychophysical tinnitus estimation or with its generation. Perhaps the simplest explanation is that lobotomy removed the emotional reinforcement of tinnitus pattern detection, removed the attentional and orienting subsystems from networks involved in the recognition of tinnitus-related activity, thus facilitating partial habituation of its signal.

Another possible mechanism of involvement of the prefrontal cortex is related to the observation that it is necessary to sustain a reaction⁷⁵. Considering the possibility of similar interaction of the prefrontal cortex with reactions of the autonomic system as with the motor system, decreased activity of the prefrontal cortex might result in decreased autonomic responses to tinnitus, whereas enhanced activity of the prefrontal cortex might produce continuous enhanced anxiety, which through a positive feedback loop might further tune the nervous system into perception of tinnitus, as discussed previously.

4.3 Parallel systems approach

The traditional hierarchical model of cortical organization postulates a stepwise hierarchical sequence, proceeding from relatively raw sensory input at the primary sensory cortices through several stages of processing, to polymodal zones for cross-modal interchange of information, to paralimbic and limbic areas for investment with emotional tone, and to the frontal association area where both sensory and limbic data are integrated⁷⁴. According to the classical model, the flow of information is mainly unidirectional, from the sensory through the associative and motor centers. Recently, Goldman-Rakic emphasized a different aspect, focusing on the functions distributed between several parallel, interconnected systems⁷⁴. She outlined the brain as a 'highly integrated but distributed machine whose resources are allocated to several basic parallel functional systems that bridge all major subdivisions of the cerebrum⁷⁴ and further stated: 'it may in the future be more useful to study the cortex in terms of information processing functions and systems rather than traditional but artificially segregated sensory, motor, or limbic components and individual neurons within only one of these components.'

This parallel distributed network concept implies the possibility of the creation of loops enhancing or suppressing the perception of a given pattern in a much more flexible manner than allowed by the hierarchical organization model, and is in accordance with Grossberg's mathematical approach to neuronal pattern recognition and classification^{22,78}. A distributed, highly interconnected network offers the possibility of creating loops which under the conditions of a continuous sensory signal (tinnitus), with significant emotional tone (concern, fear, inability to control it), further amplify the perception of tinnitus by creating positive feedback within the cortical networks.

4.4 Clinical 'masking'

On the other hand, the reason why masking and biofeedback or even a placebo are so effective in some patients is that even a partial (but sufficiently prolonged) change in the

emotional component may reorganize the functions of the distributed network, thus changing the psychological significance of the tinnitus. The loudness and pitch of tinnitus might remain the same, but the tinnitus is no longer so intrusive and can be 'ignored'.

Indeed, analysis of patients undergoing masking revealed that the psychoacoustical characteristics of tinnitus, such as its loudness or masking threshold, have not been changed either immediately after cessation of masker¹⁹⁹ nor after 6 months of its use⁹⁴. These results support the postulate that so-called 'tinnitus maskers' are effective through facilitating re-evaluation of tinnitus and retraining of cortical networks, and are not actually altering psychoacoustical characteristics of tinnitus in the direction of its attenuation.

The above postulate has obvious theoretical and clinical significance. 'Masking' should be seen as retraining of the higher processing centers involving a gradual reorganization of the recognition of tinnitus, particularly its association with emotional state. As a result, the tinnitus becomes less threatening, and there are changes in the associations of tinnitus with positive emotional states. Another important clinical implication is that during 'tinnitus masking', one should not attempt to eliminate totally the perception of tinnitus, and that a certain amount of time, measured rather in weeks than in hours, is needed for the 'masker' to be effective. The best approach is to use the 'masker' as weak as possible, which only interferes with the perception of tinnitus and provides a patient with the feeling of the possibility of even temporal control of tinnitus, without inducing anxiety by too loud a sound of 'masker'. The additional justification for the partial masking approach is that since in the 'masking procedure' behavioral retraining of the association of a sensory signal (tinnitus) with a negative emotional state is actually performed, the basic requirement for such a procedure is the presence of stimulus undergoing retraining.

Hazell^{90,91,94} showed that partial masking of tinnitus could be as effective as total or complete masking in reducing tinnitus-related complaints. It is hard to interpret this response in terms of a peripheral masking effect. It is much more likely that the masker, which in this case is simply introducing a competing signal, is facilitating retraining of tinnitus evaluation occurring at a higher cortical level. The psychological components related to the evaluation of tinnitus and the type of emotion it evokes are of particular importance, and procedures which affect this evaluation are likely to be effective in tinnitus alleviation.

5. CONCLUSIONS

Tinnitus is not a single, well-defined disease. It is a symptom of many pathologies – even in one patient, several different types of tinnitus might coexist^{83,201,202}. The approach to tinnitus presented above proposes a new view of mechanisms of its generation and perception and provides physiological interpretation of its features which have not been explained previously. Furthermore, this approach implies that tinnitus should not be simply categorized into peripheral and central, as is presently done, but that all levels are involved in each case to varying degrees, and all parts (generator, pattern recognition neural networks, and association circuits), are essential. However, dominance of a given level may exist.

Association cortices, the limbic system, and prefrontal cortex are involved in tinnitus perception and its classification and assignment of certain emotional states. Signal recognition and classification networks, which through plastic modification are able to change the recognition of a pattern of neuronal activity, are assumed to be involved in tinnitus perception. These assumptions may explain residual inhibition, rapid onset of

tinnitus but its much slower attenuation, effectiveness of contralateral masking and tinnitus masking by tones from a wide range of frequencies, its resistance to masking, and long-term effectiveness of masking tinnitus by sounds which do not 'cover' or mask it in the auditory sense.

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